Assignment_4

February 28, 2025

[ ]:
```
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```

[5]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

[14]:
```
df = pd.read_csv(r"C:\Users\hulaw\OneDrive\Desktop\DSBDA\housing.csv")
df
```

[14]:
```
        RM  LSTAT  PTRATIO      MEDV
0    6.575   4.98     15.3  504000.0
1    6.421   9.14     17.8  453600.0
2    7.185   4.03     17.8  728700.0
3    6.998   2.94     18.7  701400.0
4    7.147   5.33     18.7  760200.0
..     …      …       …        …
484  6.593   9.67     21.0  470400.0
485  6.120   9.08     21.0  432600.0
486  6.976   5.64     21.0  501900.0
487  6.794   6.48     21.0  462000.0
488  6.030   7.88     21.0  249900.0

[489 rows x 4 columns]
```

[21]:
```
x=np.array([90,80,56,76,97])
y=np.array([85,78,90,65,70])
model=np.polyfit(x,y,1)
model
```

[21]:
```
array([ -0.30831974, 102.20391517])
```

[25]:
```
predict=np.poly1d(model)
predict(65)
```

[25]:
```
82.163132137031
```

[27]:
```
y_pred=predict(x)
y_pred
```

[27]: array([74.45513866, 77.53833605, 84.93800979, 78.77161501, 72.29690049])

[29]:
```
from sklearn.metrics import r2_score
r2_score(y,y_pred)
```

[29]: 0.21927537410664832

[31]:
```
y_line=model[1]+model[0]*x
plt.plot(x,y_line,c='r')
plt.scatter(x,y_pred)
plt.scatter(x,y,c='r')
```

[31]: <matplotlib.collections.PathCollection at 0x1ec40f0e930>